Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	Yes, 3 antibodies detected, 0 RRID provided : anti-human CD63 : anti-human Calnexin antibody (YT0613 : anti-rabbit HRP Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No cell lines detected Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No organisms detected Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	The study was conducted according to the guidelines of the Declaration of Helsinki, and all procedures were approved by the Human Subjects Review Board of the First Affiliated Hospital of Anhui Medical University (number: PJ2019-06-07).	
Provide statement confirming informed consent obtained from study participants.	Materials and Methods Demographics and Samples Collection Peripheral blood samples from 6 severe burn patients and 6 healthy volunteers were obtained with the written informed consent of participants.	
Report on age and sex for all study participants.	**Age:**not detected. **Sex:**Table 1 Demographics for Participants Sampled for NGS and Patients from GSE8056 Research and Sample Type Control Group Sex Age (Years) Mean Age Burn Group Sex Age (Years) Mean Age % TBSA Burn (% Deep Burn) Mean%TBSA ABSI Score Day Postburn NGS of plasma exosome-derived small RNA Normal 01 F 24 28.3	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
	±5.28 Burn 01 F 54 39 ±16.69 60 (32) 79.83 ±19.99 11 5 Normal 02 F 21 Burn 02 M 28 60 (46) 10 5 Normal 03 F 29 Burn 03 F 25 65 (20) 11 5 Normal 04 M 28 Burn 04 M 54 97 (80) 15 4 Normal 05 M 33 Burn 05 F 54 98 (92) 16 4 Normal 06 M 35 Burn 06 M 19 99 (95) 13 4 cDNA-microarray analysis of skin GSM198875-01 F 35 38.4 ±14.37 GSM198869-01 M 8 30.8 ±14.83 6 14.67 ±12.94 No record 5 GSM198875-02 F 26 GSM198869-02 M 19 8 4 GSM198875-03 F 43 GSM198869-03 M 30 10 4 GSM198875-04 F 20 GSM198869-04 F 11 5 5 GSM198875-05 F 20 GSM198869-05 F 31 15 5 GSM198876-01 F 32 GSM198870-01 M 26 30 4 GSM198876-02 F 43 GSM198870-02 M 33 5 5 GSM198876-03 F 52 GSM198870-03 F 37 4 6 GSM198876-04 F 56 GSM198870-04 F 38 20 7 GSM198876-05 F 44 GSM198870-05 M 18 20 7 GSM198877-01 F 55 GSM198871-01 M 51 50 4 GSM198877-02 M 43 GSM198871-02 M 25 10 5 GSM198877-03 F 22 GSM198871-03 M 42 27 5 GSM198877-04 M 22 GSM198871-04 M 28 5 7 GSM198877-05 F 63 GSM198871-05 M 65 5 6 Abbreviations: TBSA, total body surface area; ABSI, abbreviate burn severity index; NGS, next-generation sequencing; F, female; M, male.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	Not detected.	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	not detected.	
Randomization	not detected.	
Blinding	not detected.	
inclusion/exclusion criteria	13 The inclusion criteria of severe burn patients included: thermal injury; % TBSA burn ≥ 60; ABSI score ≥ 10; 18 ≤ age ≤ 80; 4 ≤ days postburn ≤ 7.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Not detected.	
Define whether data describe technical or biological replicates	Not detected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	The study was conducted according to the guidelines of the Declaration of Helsinki, and all procedures were approved by the Human Subjects Review Board of the First Affiliated Hospital of Anhui Medical University (number: PJ2019-06-07).	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Burn skin specimens were sampled from a partial-thickness burned area since a full-thickness burned skin undergoing necrosis obviated the detection of gene expressions.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	18 The burn patients were clustered into 3 groups according to the post-burn days (PBD), including PBD 0–3 group (15 patients), PBD 4–7 group (15 patients) and PBD 7–17 group (15 patients), to reveal the temporal gene expression profiles in thermally injured skin.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	Not detected.	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Not detected.	
If data are publicly available, provide accession number in repository or DOI or URL.	Table 1 Demographics for Participants Sampled for NGS and Patients from GSE8056 Research and Sample Type Control Group Sex Age (Years) Mean Age Burn Group Sex Age (Years) Mean Age % TBSA Burn (% Deep Burn) Mean%TBSA ABSI Score Day Postburn NGS of plasma exosome-derived small RNA Normal 01 F 24 28.3 ±5.28 Burn 01 F 54 39 ±16.69 60 (32) 79.83 ±19.99 11 5 Normal 02 F 21 Burn 02 M 28 60 (46) 10 5 Normal 03 F 29 Burn 03 F 25 65 (20) 11 5 Normal 04 M 28 Burn 04 M 54 97 (80) 15 4 Normal 05 M 33 Burn 05 F 54 98 (92) 16 4 Normal 06 M 35 Burn 06 M 19 99 (95) 13 4 cDNA-microarray analysis of skin GSM198875-01 F 35 38.4 ±14.37 GSM198869-01 M 8 30.8 ±14.83 6 14.67 ±12.94 No record 5 GSM198875-02 F 26 GSM198869-02 M 19 8 4 GSM198875-03 F 43 GSM198869-03 M 30 10 4 GSM198875-04 F 20 GSM198869-04 F 11 5 5 GSM198875-05 F 20 GSM198869-05 F 31 15 5 GSM198876-01 F 32 GSM198870-01 M 26 30 4 GSM198876-02 F 43 GSM198870-02 M 33 5 5 GSM198876-03 F 52 GSM198870-03 F 37 4 6 GSM198876-04 F 56 GSM198870-04 F 38 20 7 GSM198876-05 F 44 GSM198870-05 M 18 20 7 GSM198877-01 F 55 GSM198871-01 M 51 50 4 GSM198877-02 M 43 GSM198871-02 M 25 10 5 GSM198877-03 F 22 GSM198871-03 M 42 27 5 GSM198877-04 M 22 GSM198871-04 M 28 5 7 GSM198877-05 F 63 GSM198871-05 M 65 5 6 Abbreviations: TBSA, total body surface area; ABSI, abbreviate burn severity index; NGS, next-generation sequencing; F, female; M, male.	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	